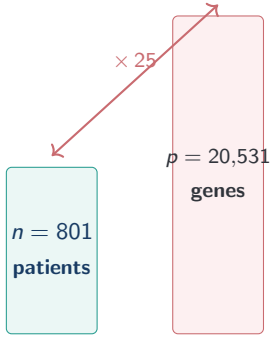
# Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks on TCGA RNA-Seq Data ($p \gg n$)

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# Why Is High-Dimensional Gene Expression Hard?

801 patients, 20,531 measured genes — a severely underdetermined system




**Statistical bottlenecks in $p \gg n$:**

- **Singular covariance:** $\mathrm{rank}(\mathbf{S}) \leq 800 \ll 20{,}531$; classical LDA fails.
- **Separability:** causes MLE divergence ($\|\mathbf{w}\|_2 \to \infty$).
- **Multicollinearity:** co-expressed genes ($|r| > 0.85$) destabilize selection.

▷ **Regularization is essential to prevent ill-posed estimation.**

SECTION 01

# Dataset and Exploratory Analysis

# TCGA RNA-Seq Dataset

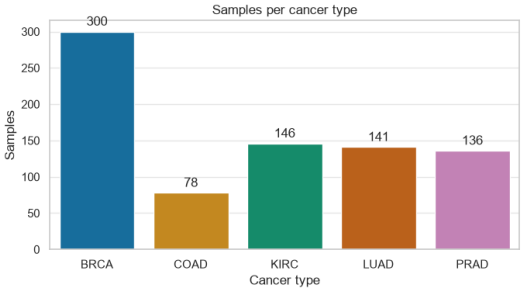
Five primary tumour types — balanced, log-transformed, pre-screened by variance

**801** samples
Total patients

**1,000** genes
MAD pre-screen

**5** classes
BRCA KIRC COAD
LUAD PRAD

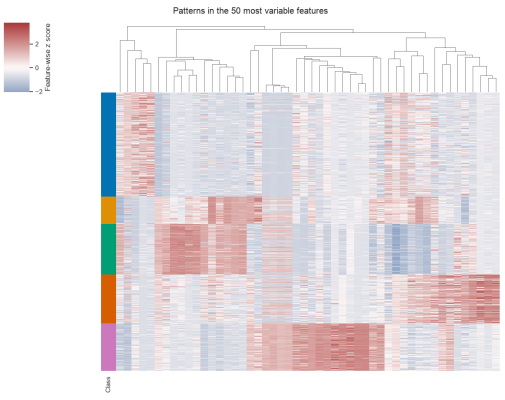
**20,531** genes
Raw transcriptome




**Preprocessing:** Log-transform $z_{ij} = \log_2(x_{ij} + 1)$ ▶ Top-1,000 by in-fold MAD (zero leakage)
▶ In-fold $Z$-standardization

# Expression Heatmap: 50 Highest-Variance Genes

Unsupervised hierarchical clustering reveals tissue-specific expression blocks



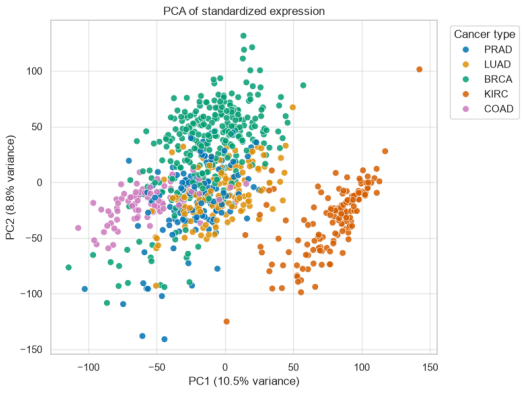

**Key observations:**

- Colours show standardised expression per gene; columns are patients.
- Cancer cohorts form **distinct blocks** without any labels.
- Row clusters reflect shared **gene modules** and pathways.
- Strong visual separation motivates linear classification.

▷ **The dataset carries strong class-specific structure even in unsupervised clustering.**

# PCA: Tumours Separate in Two Linear Dimensions

Spectral projection of 1,000 variance-filtered genes onto the leading principal components




- **PC1 (18.6%):** Separates BRCA from KIRC.
- **PC2 (12.3%):** Separates COAD and LUAD.
- **30.9%** of total variation visible in 2D.
- Five cohorts are almost fully linearly separable in this unsupervised projection.

▷ **Linear separability exists before any model fitting — non-linear methods gain little here.**

SECTION 02

# Statistical Methodology

# Three Regularized Multinomial Classifiers

Penalty geometries under collinearity

**Ridge ($L_2$)**

$\mathcal{L}_{\text{CE}} + \frac{\lambda}{2} \sum_k \|\mathbf{w}_k\|_2^2$

- Dense: **1,000 genes**.
- Best calibration (0.005).

**Lasso ($L_1$)**

$\mathcal{L}_{\text{CE}} + \lambda \sum_k \|\mathbf{w}_k\|_1$

- Sparse: **89 genes**.
- Drops collinear genes.

**Elastic Net ($L_1 + L_2$)**

$\mathcal{L}_{\text{CE}} + \lambda \sum_k [\alpha \|\mathbf{w}\|_1 + \frac{1-\alpha}{2} \|\mathbf{w}\|_2^2]$

- Grouping: **218 genes**.
- Stable feature selection.

# Why Elastic Net Handles Collinearity Better

Theorem and non-linear baselines

## Grouping Bound (Zou & Hastie)

For correlated genes ($r$):

$$\left|\hat{w}_i - \hat{w}_j\right| \leq \frac{\sqrt{2(1-r)}}{\lambda(1-\alpha)} \|\mathbf{y}\|_2$$

- $\hat{w}_i \to \hat{w}_j$ as $r \to 1$.
- Selects groups jointly.

## Non-Linear Baselines

- **Random Forest:** 200 trees.
- **XGBoost:** depth 4.
- **PCA-MLP:** 434 PCs.
- **Sparse NN:** group-$L_1$.

▷ **Trees split arbitrarily; Elastic Net groups collinear genes.**

SECTION 03

# Experimental Design

# Zero-Leakage Evaluation Pipeline

10 repeated stratified splits with strictly within-fold feature prescreening




▷ **Strict zero-leakage: feature prescreening and standardization are isolated within training folds.**

# Why Three Metrics?

Accuracy, probability calibration, and biomarker stability

## Macro-F1 (Accuracy)

$$\frac{1}{5}\sum_{k=1}^{5}\frac{2P_kR_k}{P_k+R_k}$$

Harmonic mean across classes.

**Ceiling:** all models $> 0.99$.

## Log-Loss (Calibration)

$$-\frac{1}{N}\sum_{i,k} y_{ik}\log\hat{p}_{ik}$$

Penalizes overconfidence.

**Decisive metric** for ranking.

## Stability (Biomarkers)

$$J = \frac{|S_a \cap S_b|}{|S_a \cup S_b|}$$

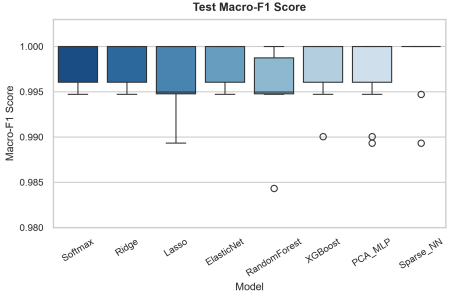
Gene overlap across splits.

**Key** for clinical biomarkers.

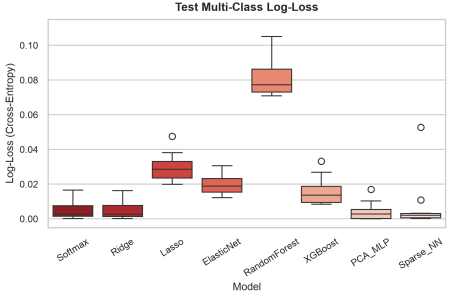
SECTION 04

# Results

# All Models Achieve High Accuracy — Log-Loss Tells the Real Story (Performance distributions across 10 repeated train/test splits)




**F1 saturates** above 0.993 for all models; accuracy alone cannot distinguish architectures.




**Log-loss diverges:** tree models (0.081) vs. regularized linear models ($< 0.020$).

# Comprehensive Model Comparison

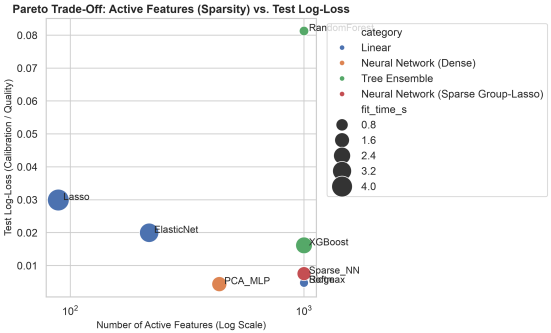
10-split cross-validated test results

| Model | Macro-F1 | Log-Loss | Genes | Jaccard |
|---|---|---|---|---|
| Softmax | 0.998 | 0.005 | 1000 | 1.00 |
| Ridge | 0.998 | 0.005 | 1000 | 1.00 |
| Lasso | 0.995 | 0.030 | **89** | 0.44 |
| **Elastic Net** | **0.998** | **0.020** | **218** | **0.57** |
| Random Forest | 0.994 | 0.081 | 1000 | 0.16 |
| XGBoost | 0.995 | 0.028 | 1000 | 0.29 |
| PCA-MLP | 0.998 | 0.004 | 434 | — |
| Sparse NN | 0.997 | 0.063 | 1000 | — |

▷ **Elastic Net matches top F1 with 218 genes and leads sparse stability.**

# Sparsity vs. Calibration: The Pareto Frontier

Elastic Net achieves optimal balance between features and calibration

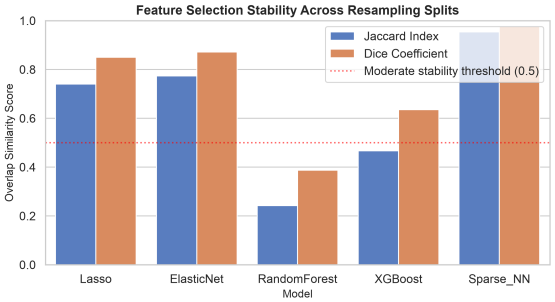



- **Ridge / PCA-MLP:** Dense (1000 genes / 434 PCs), log-loss 0.004–0.005.
- **Lasso:** Highly sparse (89 genes), log-loss 0.030.
- **Elastic Net:** Optimal trade-off (218 genes, log-loss 0.020).
- **Tree models:** Off frontier (log-loss 0.081).

▷ **Elastic Net occupies the optimal sparse Pareto frontier.**

# Feature Selection Stability

Selection consistency across 10 independent data splits

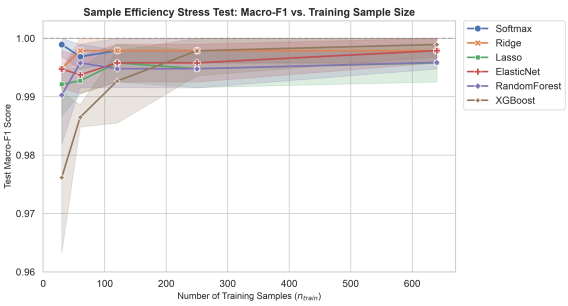



- **Elastic Net:** Jaccard 0.57 (Dice 0.73) — leads sparse models.
- **Lasso:** Jaccard 0.44 — collinear omission hurts stability.
- **XGBoost:** Jaccard 0.29 — greedy splits fluctuate.
- **Random Forest:** Jaccard **0.16** — lowest consistency.

▷ $L_2$ **grouping is essential for reproducible biomarker discovery.**

# Sample Efficiency Under Data Scarcity

Macro-F1 as training size drops from 640 to 30




- **Ridge & Elastic Net:** Maintain $F_1 \geq 0.995$ down to $n_{\text{train}} = 30$.
- **XGBoost:** Drops to 0.976 under data scarcity.
- **Random Forest:** Degrades to 0.985 at $n = 30$.

Y-axis starts at 0.96 to highlight low-$n$ divergence.

▷ **Regularized linear models remain sample-efficient; trees degrade sharply.**

SECTION 05

# Interpretation and Conclusions

# Three Main Findings

**Q1** **Flexibility provides no advantage in RNA-Seq.**

Linear separability holds. Complex models yield no F1 gain and worse calibration.

**Q2** **Collinearity destabilises Lasso and trees.**

Lasso drops genes arbitrarily; tree stability drops to 0.16. Elastic Net grouping ensures stability (0.57).

**Q3** **Elastic Net achieves optimal balance.**

Ties best F1 (0.998), selects 218 genes (78% cut), and leads in sparse stability.

# Limitations and Open Questions

## Study Limitations:

- **Single cohort:** $n = 801$; batch effects unmeasured across centres.
- **Inter-tissue task:** intra-organ subtypes remain unstudied.
- **Predictive screen:** 218 genes form statistical candidates, not a diagnostic assay.

## Non-Linear Roles:

- Subtle intra-tumour molecular subtyping.
- Higher-order epistatic gene interactions.
- Multi-modal data (genomics + pathology).

## Takeaway:

Default to Elastic Net for high-dimensional transcriptomics.

**In $p \gg n$ RNA-Seq classification,**
**regularized linear models are not a compromise —**
**they are the principled choice.**

High accuracy, sharp calibration, 78% feature reduction,
and the most reproducible biomarker sets.

Indian Statistical Institute, Kolkata | November 2026

# Thank You

Questions & Discussion